

Lecture 7: Distributions and Big Ideas In Probability

Phase 2: Risk Management Science

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Nurturing Future Risk Managers and Actuaries

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Key Definition / Result

By the end of today, you should be able to:

- ▶ Choose a **reasonable distribution model** for a real scenario involving Binomial, Poisson, or Normal distributions.
- ▶ Compute and interpret **probabilities, expected value, and standard deviation** with technology.
- ▶ Explain in words why the **Law of Large Numbers** and the **Central Limit Theorem** matter for risk management.
- ▶ Translate between **story, parameters, software output, and plain-language interpretation**.

1. Opening problem: small probabilities and many opportunities.
2. Quick refresher: random variables and distributions.
3. Part A. Revisit Binomial distribution: count of successes out of a fixed number of independent trials.
4. Part B. Poisson distribution: events per fixed time period or region.
5. Bridge. Binomial–Poisson connection: many opportunities, small chance, one useful rate.
6. Part C. Normal distribution: bell-shaped measurements, percentiles, and approximation.
7. Part D. LLN versus CLT: how averages become stable, and why Normal shapes appear.
8. Final synthesis. Why Binomial can use Normal approximation: connect Binomial counts to sums of trials and then to CLT.
9. Review sets and wrap-up.

Cognitive Trigger / Opening Problem

A micro-insurance product covers a school activity. Each participant has a small chance of a minor claim, such as a clinic visit.

Scenario.

- ▶ There are **200** participants today.
- ▶ Each participant independently files a claim with probability **1%**.
- ▶ The organiser worries: “Will we see **0**, **a few**, or **many** claims?”

Investigation / Activity: Motivating Questions

- ▶ What is the probability of **no claim at all**?
- ▶ What is the probability of **more than 5 claims**?
- ▶ If we repeat similar events many times, what happens to the **average number of claims**?

Key Definition / Result

This same story can be modelled in **two** ways:

- ▶ **Binomial model:** count successes out of n trials. This is the exact fixed-trial model.
- ▶ **Poisson model:** approximate rare events using the rate $\lambda = np$. This is useful when n is large and p is small.

Technology Check

At the end of today, we will explain why a Binomial count can sometimes be approximated by a Normal curve.

Pause-and-Think

If two models give similar numbers, does that mean they are equally good explanations of the world?

Key Definition / Result

A **random variable** X is a numerical summary of an uncertain outcome.

- ▶ A **discrete random variable** takes countable values such as $0, 1, 2, \dots$. We use a probability mass function $P(X = x)$.
- ▶ A **continuous random variable** takes values on an interval. We use a probability density function $f(x)$, and probabilities come from **area**.

Investigation / Activity: Risk-Management Lens

Counts of claims are usually discrete. Sizes of losses are often continuous.

Key Definition / Result

- ▶ $E[X]$, or the mean, is approximately the **long-run average** value of X .
- ▶ $SD(X)$ measures the **typical variation** around the mean.

Investigation / Activity: Interpretation Sentence

Always report probabilities with a sentence, for example:

"The probability is 0.12, meaning about 12 out of 100 similar days would look like this under the model."

Technology Check: R Function Map

- ▶ Binomial: `dbinom(x, size, prob)` and `pbinom(x, size, prob)`.
- ▶ Poisson: `dpois(x, lambda)` and `ppois(x, lambda)`.
- ▶ Normal: `pnorm(x, mean, sd)` and `qnorm(q, mean, sd)`.

Part A. Binomial Distribution

Structural Core: When Does Binomial Appear?

The Binomial distribution appears whenever we count how many times something happens across a **fixed** number of trials:

- ▶ the number of claims among n policyholders,
- ▶ the number of defective items in a batch of n ,
- ▶ the number of students who answer a question correctly out of n .

Technology Check

Today, focus on **assumptions**, **interpretation**, and **technology**.

Key Definition / Result

We write $X \sim \text{Bin}(n, p)$ if:

- ▶ there are n trials,
- ▶ each trial is **success/failure**,
- ▶ the success probability is p each time,
- ▶ trials are **independent**.

Then X counts the number of successes.

Structural Core: Parameter Meaning

n is the number of opportunities. p is the chance per opportunity.

Pause-and-Think

Which assumption is easiest to state but hardest to justify in a real insurance setting?

Key Definition / Result

For $X \sim \text{Bin}(n, p)$,

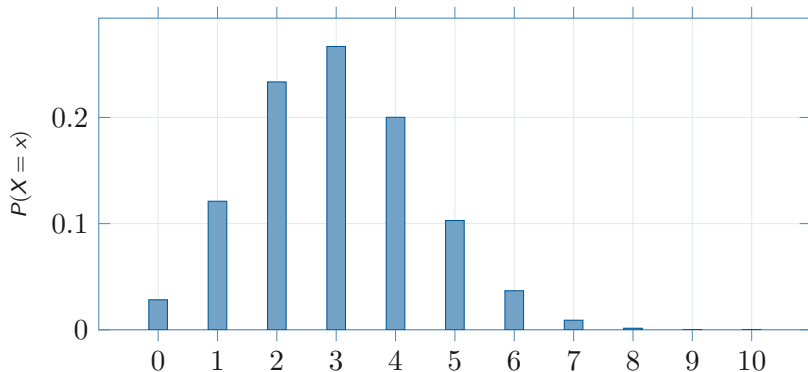
$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n.$$

$$E[X] = np, \quad \text{Var}(X) = np(1 - p), \quad \text{SD}(X) = \sqrt{np(1 - p)}.$$

Technology Check

In labs, you will see [how](#) these formulas arise. Today we mainly [use](#) them and interpret them in risk language.

Example: $X \sim \text{Bin}(10, 0.3)$



Investigation / Activity

The shape depends on p . If p is small, the mass concentrates near 0.

Worked Example

A factory tests $n = 20$ items. Historically, each item is defective with probability $p = 0.05$, assuming independence.

1. Find $P(X \leq 2)$, where X is the number of defectives.
 2. Find the expected number of defectives and the standard deviation.
-
1. Identify the model: $X \sim \text{Bin}(20, 0.05)$.
 2. Use technology for $P(X \leq 2)$, because this is a cumulative probability.
 3. Compute $E[X] = np$ and $\text{SD}(X) = \sqrt{np(1 - p)}$.
 4. Translate the numerical answer into a sentence about repeated batches.

Technology Check: R Workflow

```
1 # P(X <= 2) for Binomial(20, 0.05)
2 number_of_items <- 20
3 defect_probability <- 0.05
4
5 pbinom(2, size = number_of_items, prob = defect_probability)
6
7 expected_defectives <- number_of_items * defect_probability
8 sd_defectives <- sqrt(
9   number_of_items * defect_probability * (1 - defect_probability)
10 )
11
12 expected_defectives
13 sd_defectives
```

Investigation / Activity: Interpretation Sentence

If the output is about 0.92, then about **92 out of 100 similar batches** would have at most 2 defectives under this model.

Worked Example

An insurer covers $n = 500$ similar low-risk policyholders for one week. Each policyholder files a claim with probability $p = 0.004$, assuming independence.

1. What is the expected number of claims?
 2. What is the standard deviation?
 3. Give a **plain-language** interpretation.
-
1. Model the number of claims as $X \sim \text{Bin}(500, 0.004)$.
 2. Compute the mean $E[X] = np$.
 3. Compute the standard deviation $\sqrt{np(1 - p)}$.
 4. Interpret the standard deviation as typical variation around the mean.

Common Mistake / Pitfall

Binomial is **not** appropriate if:

- ▶ p is **not constant**, because risk differs across people;
- ▶ trials are **dependent**, because one claim may trigger more claims;
- ▶ the number of trials is **not fixed**, because exposure varies.

Deepening Layer: Fix Mindset

If assumptions fail, you may need stratification with different probabilities, time-varying models, or a different distribution.

Pause-and-Think

Which failure is most likely in real insurance: non-constant probability, dependence, or changing exposure?

Investigation / Activity

Use technology to explore:

1. Simulate 10,000 weeks of $X \sim \text{Bin}(500, 0.004)$.
2. Plot a histogram of simulated X .
3. Compare the histogram to theoretical probability mass function values.
4. Write **two sentences**: what do you notice about the typical size and spread?

```
1 simulated_claim_counts <- rbinom(  
2   n = 10000,  
3   size = 500,  
4   prob = 0.004  
5 )  
6 mean(simulated_claim_counts)  
7 sd(simulated_claim_counts)  
8 hist(  
9   simulated_claim_counts,  
10  breaks = 40,  
11  col = "gray",  
12  main = "Simulated claim counts"  
13 )
```

Investigation / Activity

Answer quickly without a calculator first, then verify with technology:

1. If $X \sim \text{Bin}(n, p)$, what does np mean in the story?
2. When would independence fail in a real insurance setting?
3. Which function gives $P(X \leq k)$ in R for a Binomial distribution?

Technology Check: Verification In R

```
1 # Template:  
2 pbinom(k, size = n, prob = p)
```

Part B. Poisson Distribution

Structural Core: Classic Use-Cases

- ▶ Number of claims arriving per day.
- ▶ Number of website errors per hour.
- ▶ Number of accidents at an intersection per month.

Investigation / Activity: Actuarial Lens

One parameter λ often gives a surprisingly useful first model.

Key Definition / Result

We write $X \sim \text{Pois}(\lambda)$ if X counts events with average rate λ in a fixed period or region.

$$P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

$$E[X] = \lambda, \quad \text{Var}(X) = \lambda, \quad \text{SD}(X) = \sqrt{\lambda}.$$

Pause-and-Think

Why is it unusual, and useful, that the Poisson mean and variance are equal?

Key Definition / Result

If n is large and p is small, with $\lambda = np$, then

$$\text{Bin}(n, p) \approx \text{Pois}(\lambda).$$

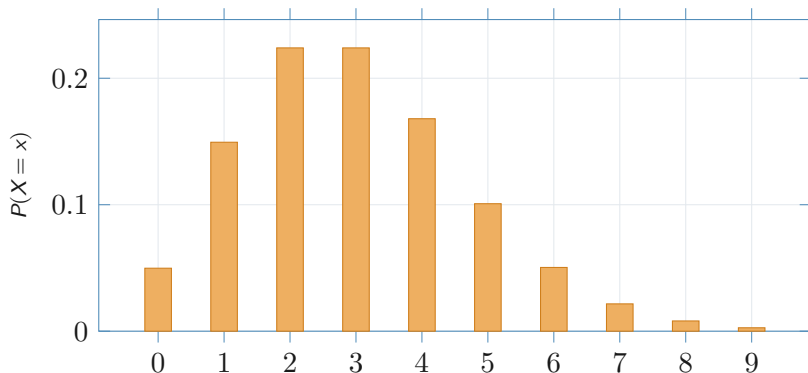
Investigation / Activity: The Connection

When n is very large (many opportunities) and p is very small (rare events), the Binomial distribution perfectly transforms into the Poisson distribution. Instead of juggling two parameters (n and p), we can approximate the risk using a single arrival rate: $\lambda = np$.

Deepening Layer: Why This Matters

It simplifies computation when n is huge and we only care about a small count. This connects directly to the opening problem: $n = 200$, $p = 0.01$, so $\lambda = 2$.

Example: $X \sim \text{Pois}(3)$



Investigation / Activity

As λ increases, the Poisson distribution becomes more symmetric and later looks more Normal-like.

Worked Example

A small insurer observes an average of $\lambda = 1.8$ claims per day. Assume $X \sim \text{Pois}(1.8)$ for today's claim count.

1. Find $P(X = 0)$, a quiet day.
 2. Find $P(X \leq 2)$, at most two claims.
1. Identify the model and parameter λ .
 2. Use `dpois` for exact probability and `ppois` for cumulative probability.
 3. Interpret the result in context.

Technology Check: R Workflow

```
1 daily_claim_rate <- 1.8
2
3 dpois(0, lambda = daily_claim_rate) # P(X = 0)
4 ppois(2, lambda = daily_claim_rate) # P(X <= 2)
```

Investigation / Activity: Interpretation Sentence

If $P(X = 0) = 0.165$, then about 16–17 out of 100 days would have zero claims under this model.

Worked Example

In 30 days, a department recorded 54 incoming claims.

1. Estimate the daily rate λ .
2. Use your model to estimate $P(X \geq 4)$ for tomorrow.

1. Estimate the rate:

$$\hat{\lambda} = \frac{54}{30} \text{ claims per day.}$$

2. Compute $P(X \geq 4) = 1 - P(X \leq 3)$ using `ppois`.
3. Translate the probability into a planning statement.

Pause-and-Think

If claims arrive in clusters, would the Poisson model under-estimate or over-estimate the chance of an extreme day?

Technology Check: R Workflow

```
1 total_claims <- 54
2 number_of_days <- 30
3
4 estimated_lambda <- total_claims / number_of_days
5
6 # P(X >= 4) = 1 - P(X <= 3)
7 1 - ppois(3, lambda = estimated_lambda)
```

Common Mistake / Pitfall

Do not treat $\hat{\lambda}$ as a perfect truth. It is an estimate from limited data.

Common Mistake / Pitfall

Poisson may be inappropriate if:

- ▶ events cluster together;
- ▶ the rate changes over time;
- ▶ different subgroups have very different risk levels;
- ▶ the variance is much larger than the mean.

Working Example

A stable call centre with a roughly constant average number of calls per hour may be a reasonable first Poisson example.

Boundary / Failing Example

A storm day that causes many related insurance claims is not well represented by independent arrivals at a constant rate.

Investigation / Activity: Boundary Condition

- ▶ **Critical condition highlighted.** Poisson needs a reasonably stable rate and limited clustering.
- ▶ **Thinking direction.** When the real world is more variable than the model, check whether the model is under-estimating tail risk.

AI Exploration: Google Gemini Exploration

For a deeper dive, try this with Google Gemini:

Explain how the shape of a Poisson distribution changes when λ increases from 1 to 3 to 10. Use one insurance example and one school-life example. Then identify one situation in which the Poisson model may fail because events arrive in clusters.

Investigation / Activity

Use technology to compare Poisson distributions with $\lambda = 1$, $\lambda = 3$, and $\lambda = 10$. Describe how the centre, spread, and symmetry change.

```
1 x_values <- 0:25
2
3 plot(
4   x_values,
5   dpois(x_values, lambda = 1),
6   type = "h",
7   lwd = 2,
8   ylim = c(0, 0.4),
9   xlab = "Number of events",
10  ylab = "Probability"
11 )
12
13 points(x_values, dpois(x_values, lambda = 3), type = "h", col = "blue")
14 points(x_values, dpois(x_values, lambda = 10), type = "h", col = "red")
```

Part C. Normal Distribution

Structural Core

The Normal distribution is useful for modelling quantities that vary around a central value with roughly symmetric deviations.

- ▶ Delivery time may vary around an average time.
- ▶ Measurement error may vary around a true value.
- ▶ Portfolio returns are sometimes approximated by Normal models, although this must be checked carefully.

Pause-and-Think

Why might a symmetric bell curve be attractive mathematically but dangerous if the real data have extreme losses?

Key Definition / Result

We write

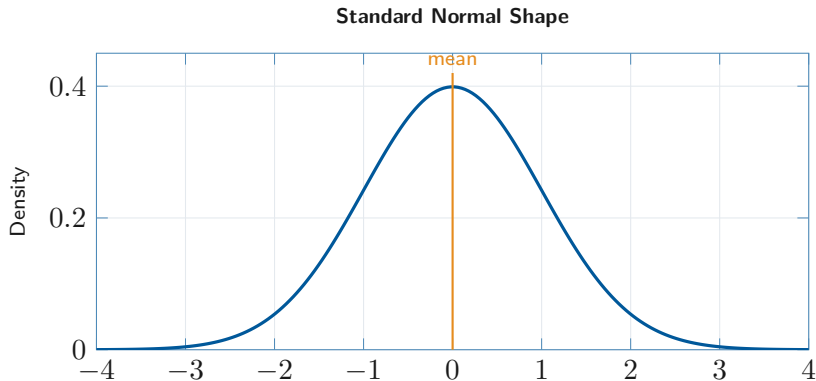
$$X \sim \mathcal{N}(\mu, \sigma^2)$$

when X follows a Normal distribution with mean μ and variance σ^2 .

The standard deviation is σ , not σ^2 .

Investigation / Activity: Interpretation Sentence

μ controls the centre. σ controls the typical spread around the centre.



Investigation / Activity

The total area under the curve is 1. Probabilities are areas under the curve.

Worked Example

Delivery time T is modelled as

$$T \sim \mathcal{N}(32, 6^2),$$

where T is measured in minutes. Find $P(T \leq 40)$.

1. Identify the mean $\mu = 32$ and standard deviation $\sigma = 6$.
2. Use `pnorm(40, mean=32, sd=6)`.
3. Interpret the output as the proportion of similar deliveries that would take no more than 40 minutes.

Technology Check: R Workflow

```
1 delivery_mean <- 32
2 delivery_sd <- 6
3
4 pnorm(40, mean = delivery_mean, sd = delivery_sd)
5 pnorm(45, mean = delivery_mean, sd = delivery_sd) -
6   pnorm(25, mean = delivery_mean, sd = delivery_sd)
```

Investigation / Activity: Interpretation Sentence

A Normal probability should always be interpreted as an area, not as a probability at one exact point.

Worked Example

Delivery time $T \sim \mathcal{N}(32, 6^2)$. What time threshold t is the **90th percentile**, meaning $P(T \leq t) = 0.90$?

1. Recognise this as an **inverse cumulative distribution function** problem.
2. Use `qnorm(0.90, mean=32, sd=6)`.
3. Interpret the result: 90% of deliveries are **no more than** t minutes under the model.

Technology Check: R Workflow

```
1 delivery_mean <- 32
2 delivery_sd <- 6
3
4 qnorm(0.90, mean = delivery_mean, sd = delivery_sd)
5 qnorm(0.95, mean = delivery_mean, sd = delivery_sd)
```

Pause-and-Think

How is a percentile different from a probability? Which direction does the function move: value to probability, or probability to value?

Common Mistake / Pitfall

- ▶ σ is the **standard deviation**, with the same unit as the data.
- ▶ σ^2 is the **variance**, with squared units.
- ▶ A model written as $\mathcal{N}(\mu, \sigma^2)$ uses **variance** in the notation.

Technology Check

Always label your parameters with units: minutes, dollars, centimetres, or other relevant units.

Investigation / Activity

Pick a dataset, such as classroom measurements or a provided sample. Use technology to:

1. plot a histogram and overlay a Normal curve using the sample mean and standard deviation;
2. write **two sentences**: does Normal seem plausible, and where does it fail?

```
1 sample_values <- rnorm(200, mean = 32, sd = 6) # placeholder dataset
2
3 hist(sample_values, breaks = 20, col = "gray")
```


Key Definition / Result

Normal is not only a model. It is also an approximation tool:

- ▶ Binomial with large n and moderate p .
- ▶ Poisson with large λ .
- ▶ Sums and averages through the Central Limit Theorem.

Deepening Layer: Practical Advice

Always verify with technology by comparing exact and approximate answers when possible.

Part D. Law of Large Numbers and Central Limit Theorem



Key Definition / Result

Law of Large Numbers, informal version: If we repeat an experiment many times, the **sample average** tends to get closer to the **true mean**.

Investigation / Activity: Risk-Management Lens

If average claim cost is μ , then observing more claims makes our estimate of μ **more stable**. The theorem is powerful because it converts uncertainty at the individual level into predictability at the portfolio level: one person's claim is hard to predict, but the average behaviour of many similar risks can become surprisingly reliable.

Deepening Layer: Young Learner Intuition

Think of repeatedly tossing a fair coin. After 10 tosses, the proportion of heads may look quite far from 0.5. After 10,000 tosses, the proportion usually looks much closer to 0.5. The result does **not** say the next few tosses must compensate for earlier imbalance; it says the long-run average becomes less sensitive to any one unusual observation.

Pause-and-Think

Does the Law of Large Numbers say that short runs must balance out immediately?

Deepening Layer: Plain Language

Small samples are noisy; big samples tend to settle down.

Investigation / Activity: Risk Management Lens

If an insurance company covers only 5 people, their average cost fluctuates wildly. If they cover 5,000 people, the average claim amount per person stabilizes near the true theoretical mean.

Technology Check: Simulation Demonstration

- ▶ Generate many observations.
- ▶ Track the running mean.
- ▶ Watch it stabilise.

```
1 simulated_counts <- rpois(5000, lambda = 2)
2 running_mean <- cumsum(simulated_counts) /
3   seq_along(simulated_counts)
4 plot(
5   running_mean,
6   type = "l",
7   col = "blue",
8   main = "LLN: running mean stabilises"
9 )
10 abline(h = 2, col = "red")
```

Why The Law of Large Numbers Is So Useful I

Key Definition / Result

The Law of Large Numbers tells us that a carefully collected average from many observations can be a meaningful estimate of an underlying quantity.

Structural Core: Examples

- ▶ **Insurance:** one policyholder is unpredictable, but average loss per policyholder becomes more stable when many similar policyholders are pooled.
- ▶ **Data science:** a model trained or evaluated with too little data may behave erratically, while larger representative samples give more trustworthy estimates.

Common Mistake / Pitfall

LLN does **not** promise that every individual outcome becomes predictable. It only says that the **average** of many observations becomes more stable under suitable conditions.

Key Definition / Result

Central Limit Theorem, informal version: For many independent observations with finite variance, the **sample mean** is approximately Normal when the sample size is large.

Deepening Layer: What This Enables

The Central Limit Theorem lets us approximate probabilities about averages and totals without knowing the exact distribution. This is much stronger than just saying *averages settle down*: it tells us the **shape** of the remaining uncertainty around the mean.

Investigation / Activity: Why Young Actuaries Should Care

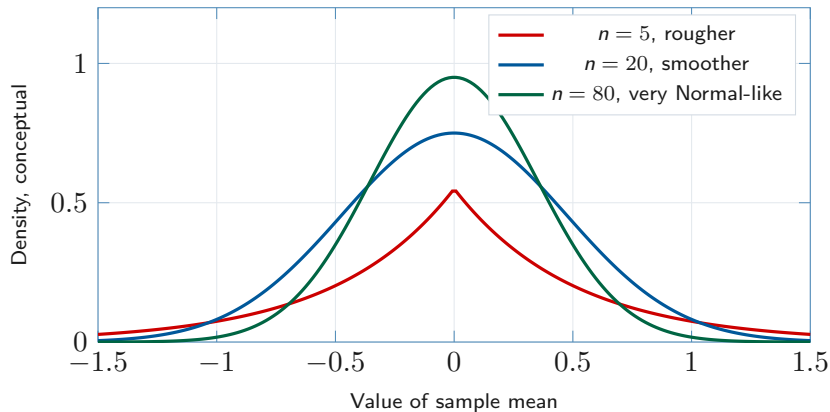
Even when individual data are skewed, irregular, or not bell-shaped, the average or total of many independent observations often behaves in a bell-shaped way after proper scaling. That is why Normal approximations appear in confidence intervals, hypothesis tests, pricing approximations, reserve calculations, quality control, and many simulation summaries.

Technology Check

We use the Central Limit Theorem as a **tool** first. Proof is for later.

Central Limit Theorem: Visual Intuition I

As sample size increases, the distribution of the mean becomes more Normal-like



Investigation / Activity

The point is the **shape trend**, not the exact formula in this conceptual picture.

Deepening Layer: Risk Management Connection

While individual insurance claims are often heavily skewed (not Normal at all), the **average** of many independent claims becomes remarkably Normal-like (bell-shaped). This is why the Normal curve is the ultimate approximation tool in risk management.

Key Definition / Result

Both the Law of Large Numbers and the Central Limit Theorem talk about what happens when the sample size becomes large, but they answer **different** questions.

	Law of Large Numbers	Central Limit Theorem
Main question	Does the sample average settle near the true mean?	What is the approximate shape of the sample average?
Main message	The average becomes stable .	The average becomes Normal-like .
Object of interest	$\bar{X}$ getting close to μ .	The distribution of $\bar{X}$, or of a standardised version of $\bar{X}$.
Risk interpretation	Large portfolios make average loss more predictable.	Large portfolios allow probability calculations for totals and averages.

Cognitive Trigger / Opening Problem

Imagine many students each make a small random prediction error when estimating the time needed to finish a task. One student's error may be large, positive, negative, or unusual. But if we average many independent students' errors, two things happen.

1. **LLN**: the average prediction error tends to move closer to the true average error. Random noise is diluted by many observations.
2. **CLT**: the distribution of that average error often becomes bell-shaped, so we can estimate how likely it is that the average error exceeds a certain threshold.
3. **Practical meaning**: LLN gives confidence that a large average is meaningful; CLT gives a calculator-friendly way to measure uncertainty around that average.

Pause-and-Think

If LLN says the average becomes stable, why do we still need CLT to make decisions under risk?

Worked Example

Daily claim counts have mean $\mu = 2$ and variance $\sigma^2 = 2$, as in a Poisson-like setting. Let $\bar{X}$ be the average claim count over $n = 49$ days. Approximate

$$P(\bar{X} \geq 2.4).$$

1. By the Central Limit Theorem, $\bar{X}$ is approximately Normal with mean μ and standard deviation $\sigma/\sqrt{n}$.
2. Compute the Z-score:

$$z = \frac{2.4 - \mu}{\sigma/\sqrt{n}}.$$

3. Convert the Z-score to probability using `pnorm`.

Technology Check: R Workflow

```
1 claim_mean <- 2
2 claim_variance <- 2
3 number_of_days <- 49
4 claim_sd <- sqrt(claim_variance)
5 standard_error <- claim_sd / sqrt(number_of_days)
6 z_score <- (2.4 - claim_mean) / standard_error
7 1 - pnorm(z_score)
```

Investigation / Activity: Interpretation Sentence

The answer is an approximate probability that the **average** daily claim count over 49 days is at least 2.4. The larger the sample size, the smaller the standard error $\sigma/\sqrt{n}$, so unusually high averages become less likely if the model assumptions are reasonable.

Common Mistake / Pitfall

Law of Large Numbers and Central Limit Theorem intuition may fail or converge slowly when:

- ▶ observations are **dependent**, such as clusters or contagion;
- ▶ variance is huge or the tail is heavy, as in extreme losses;
- ▶ the sample is not representative, causing bias.

Investigation / Activity: Actuarial Lens

When stakes are high, always stress-test assumptions and use diagnostics together with scenario thinking.

Pause-and-Think

Why could a mathematically correct theorem still lead to a bad decision?

Key Definition / Result

A Binomial random variable is not mysterious: it is a **sum of many 0–1 trials**. If $X \sim \text{Bin}(n, p)$, then we can write

$$X = I_1 + I_2 + \cdots + I_n,$$

where each I_i equals 1 for success and 0 for failure.

Deepening Layer: CLT Connection

When n is large and the success probability is not too close to 0 or 1, the sum of many independent 0–1 variables behaves approximately like a Normal distribution. This is the Central Limit Theorem working behind the scenes.

Investigation / Activity: Why Not Always?

If p is very small, most trials are failures and the distribution may remain strongly right-skewed. In that rare-event case, the Poisson approximation is often more natural. This is why today's flow matters: Binomial first, Poisson for rare events, Normal for large balanced sums, CLT as the big explanation.

Key Definition / Result

If n is large and p is not too extreme, then

$$X \sim \text{Bin}(n, p) \approx \mathcal{N}(np, np(1 - p)).$$

Investigation / Activity: The Connection

If n is large and p is moderate, the discrete “staircase” of Binomial counts blends perfectly into the continuous, smooth bell shape of the Normal distribution. This allows us to use continuous area as a mathematical shortcut for summing many discrete probabilities.

Deepening Layer: What We Do Today

- ▶ Use approximation to compute probabilities quickly.
- ▶ Mention **continuity correction** as a practical tweak, with details moved to tutorial.

Worked Example

A platform sends $n = 200$ notifications. Each is opened with probability $p = 0.02$. Approximate $P(X \geq 8)$, where $X \sim \text{Bin}(200, 0.02)$.

1. Compute the mean and standard deviation:

$$\mu = np, \quad \sigma = \sqrt{np(1-p)}.$$

2. Convert $X \geq 8$ to a Normal probability, optionally with continuity correction.
3. Use `pnorm` or a spreadsheet Normal cumulative distribution function.

Technology Check

In tutorial, compare the exact `pbinom` answer against the Normal approximation.

Technology Check: R Workflow

```
1 notification_count <- 200
2 open_probability <- 0.02
3 exact_probability <- 1 - pbinom(7, size = notification_count, prob = open_probability)
4 normal_mean <- notification_count * open_probability
5 normal_sd <- sqrt(notification_count * open_probability * (1 - open_probability))
6 # Continuity correction: P(X >= 8) becomes P(Y >= 7.5)
7 normal_approximation <- 1 - pnorm(7.5, mean = normal_mean, sd = normal_sd)
8 exact_probability
9 normal_approximation
```

Investigation / Activity: What Students Should Notice

The exact Binomial calculation is the benchmark when technology is available. The Normal approximation is a powerful shortcut and a conceptual bridge, but students should learn to compare it against the exact answer before trusting it blindly.

Investigation / Activity: Instructions

Use technology for computation, but write **one interpretation sentence** for each answer.

1. Identify whether the scenario is Binomial, Poisson, Normal, or none: 10 coin tosses; number of calls per hour; heights of students.
2. If $X \sim \text{Bin}(40, 0.1)$, find $E[X]$ and the standard deviation.
3. Compute $P(X \leq 3)$ for $X \sim \text{Bin}(40, 0.1)$.
4. If $Y \sim \text{Pois}(2.5)$, compute $P(Y = 0)$ and $P(Y \geq 4)$.
5. If $Z \sim \mathcal{N}(100, 15^2)$, compute $P(Z > 130)$.
6. Find the 95th percentile of $\mathcal{N}(100, 15^2)$.
7. Explain the Law of Large Numbers in one sentence using the idea of average claim count.
8. Explain the Central Limit Theorem in one sentence using the idea of average.

Investigation / Activity: Instructions

Show your modelling choice and parameter setup clearly.

1. Opening problem: $n = 200$, $p = 0.01$. Compute $P(X = 0)$ exactly and by Poisson approximation; comment on the gap.
2. For $X \sim \text{Bin}(300, 0.03)$, approximate $P(X \geq 15)$ using Normal approximation and mention the continuity-correction idea.
3. A system has 3 errors per day on average. If you observe 10 days, what is the probability that total errors exceed 40? Hint: use Poisson additivity.
4. If $T \sim \mathcal{N}(32, 6^2)$, find $P(25 \leq T \leq 45)$ and interpret it.
5. Use the Central Limit Theorem to approximate $P(\bar{X} \leq 1.7)$ for $n = 100$ days when the daily mean is 2 and the variance is 2.
6. Explain one realistic reason why the Poisson model may under-estimate variability in real claims.
7. Design a quick simulation to check a modelling assumption you made above.
8. Reflection: which tool helped you most today, R, spreadsheet, or Google Gemini? Why?

Key Definition / Result

Today's takeaway:

- ▶ **Binomial** means count of successes out of n independent trials.
- ▶ **Poisson** means events per period with rate λ , and it can also act as a rare-event approximation when n is large and p is small.
- ▶ **Normal** is both a bell-curve model and an approximation tool for large sums and averages.
- ▶ **Law of Large Numbers** explains why averages stabilise.
- ▶ **Central Limit Theorem** explains why averages often become Normal-like and why Binomial counts can sometimes use Normal approximation.

End Of Lecture 7